



REVA Academy for Corporate Excellence

Program: MTech/MSc in Artificial Intelligence	Course: Text Analytics - NLP
Batch: AI 09	Exam: Final
Date: 19 th July 2025	Time: 10:00am to 1:00pm
Duration: 3 Hrs	Marks: 100

Instructions

1.	This is an open book test, GenAI Tools are allowed, but limit your responses to the point, do not insert all garbage spit out by Gen AI tool!
2.	To the point answers will get marks, unnecessary text answers will be penalized.
3.	Be careful about using Gen AI tools to answer the questions.
4.	All the artefacts must be zipped and submitted through the LMS Link. The Zipped Folder must be named as 'SRN_Name of the Course'.

Instructions		Answer ONE FULL question from each unit.	Marks
UNIT - I			
Q1.	a.	Define tokenization in NLP. Give an example.	25
	b.	What is a <i>Bag of Words</i> model? How does <i>count vectorization</i> differ from <i>one-hot encoding</i> ?	
	c.	Explain the purpose of <i>TF-IDF</i> . When is it preferred over simple frequency-based representations?	
	d.	What are <i>stop words</i> , and why are they removed in text pre-processing?	
	e.	List any four challenges commonly encountered during text pre-processing.	
	f.	Explain the process and benefits of using <i>Word2Vec</i> embeddings over traditional vectorization techniques.	
	g.	How does <i>semantic analysis</i> differ from <i>sentiment analysis</i> ? Illustrate with a use case.	
	h.	Explain the concept of <i>n-gram frequency analysis</i> using unigrams and bigrams. Provide an example from a sample sentence.	
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Q2.	a	Describe the typical NLP project pipeline with examples of each stage.	25
	b	You are provided with the following text: "I absolutely love this phone! 🤖 It's super-fast, and the battery life is amazing!!!" i. Write a Python function to clean this text (remove punctuation, lowercase, remove stop words). ii. Apply tokenization, stemming, and lemmatization (use any <i>dummy stemmer/lemmatizer</i> logic or library-based function). iii. Generate its unigram frequency distribution using any Python library (<i>NLTK</i>, <i>spaCy</i>, or manually).	
	c	Imagine you're building a review analysis system for an e-commerce website. i. Describe how you would use web scraping to collect review data. ii. Explain the pipeline you would create to process the collected reviews. iii. How would you store and visualize the processed text data for actionable insights?	
	d.	What does <i>TF</i> and <i>IDF</i> stand for in the TF-IDF model? How does each component contribute to final weighting?	
UNIT - II			
Q3.	a.	Differentiate between rule-based and data-driven text classification approaches.	25
	b.	What are semantic rules in the context of rule-based classifiers? Give an example.	
	c.	What is the purpose of <i>AFINN lexicon</i> in sentiment analysis?	
	d.	Explain the concept of <i>polarity strength</i> at the sentence level with an example.	
	e.	What is a spam classifier? Mention any two features useful in building one.	
	f.	How does <i>Naïve Bayes</i> handle the text classification problem using probabilities?	
	g.	What role does corpus creation play in building a sentiment analysis model?	
	h.	Describe how you would apply a rule-based lexicon approach using NRC or custom lexicons .	

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	i.	How would you enhance your model to handle <i>sarcasm or negations</i> ?	
	j.	Suppose the client wants polarity at paragraph-level granularity. What design decisions will you make?	
	OR		
Q4.	a.	Compare AFINN , NRC , and a Custom lexicon in terms of sentiment tagging capabilities.	25
	b.	Explain how a score-based classifier using lexicons works. Describe with a pseudo-code or step-by-step approach.	
	c.	Describe the workflow for sentiment analysis using an SVM classifier .	
	d.	Explain how polarity is computed at paragraph level using lexicon-based methods.	
	e.	Discuss the key steps in building a spam classification pipeline using supervised learning.	
	f.	In the context of the IMDB Movie Dataset , explain why SVM might outperform Naïve Bayes in certain cases	
	g.	Write Python code to load the data (e.g., using <code>keras.datasets.imdb.load_data()</code> or directly from the raw files), then apply text preprocessing (cleaning, tokenization, stopword removal).	
	h.	Build a Naïve Bayes classifier using scikit-learn to predict review sentiment.	
	UNIT - III		
Q5.	a.	Briefly explain cosine similarity and its use in document similarity search.	25
	b.	What are the core components of a Transformer architecture?	
	c.	How does SVD help in reducing dimensionality in NLP tasks?	
	d.	Mention two challenges in unsupervised topic detection.	
	e.	Explain how document clustering is applied in real-world search engines or digital libraries. Include the role of feature vectors.	

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	f.	Compare and contrast TF-IDF + KMeans vs LSA (SVD) + KMeans for topic modeling.	
	g.	Describe a typical pipeline for extractive text summarization using traditional methods (e.g., TextRank).	
	h.	Use the “20 Newsgroups Text Dataset” from scikit-learn or Kaggle. 1. Load and preprocess the documents (tokenization, TF-IDF). 2. Apply Truncated SVD (LSA) to reduce dimensions. 3. Perform KMeans clustering on the reduced feature set. 4. Visualize clusters using t-SNE or matplotlib (optional but bonus-worthy). 5. Evaluate how meaningful the clusters are based on sample documents. <i>Dataset:</i> 20 Newsgroups Dataset – available via <code>sklearn.datasets.fetch_20newsgroups</code>.	
	OR		
Q6.	a.	What is the difference between document clustering and topic modeling?	25
	b.	Define Latent Semantic Analysis (LSA) and explain its role in topic modeling.	
	c.	What are the key differences between extractive and abstractive summarization?	
	d.	What is the role of the Transformers Pipeline in modern NLP summarization tasks? List common models used.	
	e.	Explain how semantic topic detection can be achieved using LSA with a working example (no code required). Use any article from the CNN/DailyMail or Wikipedia dataset.	
	f.	Implement extractive summarization using TextRank from gensim or sumy.	
	g.	Implement abstractive summarization using Hugging Face transformers (e.g., facebook/bart-large-cnn).	
	h.	<i>Dataset:</i> CNN/DailyMail summarization dataset via Hugging Face: <code>cnn_dailymail</code> https://huggingface.co/datasets/cnn_dailymail	
UNIT - IV			

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Q7.	a.	What are Large Language Models (LLMs) ? How are they different from traditional NLP models?	25
	b.	Mention two use cases each of ChatGPT and LLAMA3 in enterprise settings.	
	c.	Define prompt engineering and explain why it is important for LLMs.	
	d.	Explain the architecture and training objectives behind GPT-style decoder-only LLMs	
	e.	Compare and contrast ChatGPT (GPT-4) with Mistral or Mixtral in terms of architecture, licensing, and usability	
	f.	Describe the key components of a RAG system and how it improves over zero-shot generation.	
OR			
Q8.	a.	What is a retriever in a RAG (Retrieval-Augmented Generation) system?	25
	b.	Name two open-source LLMs and state one advantage of using each.	
	c.	What tasks in NLP can be automated using LLMs beyond summarization and question answering?	
	d.	How does Mixtral (Mixture of Experts) differ from traditional transformer-based LLMs? Describe the key components of a RAG system and how it improves over zero-shot generation.	
	e.	How would you design a text classification pipeline using a pre-trained LLM (like <i>BERT or LLAMA3</i>)?	
	f.	What are the challenges in fine-tuning large open-source models on a domain-specific corpus?	
